Experiment 6: Create a Scatter Plot of the Trading Volume and Stock Prices of Alphabet Inc. Between Two Specific Dates

Aim

To write a Python Pandas program to create a **scatter plot** showing the relationship between the **trading volume** and **closing stock prices** of Alphabet Inc. between two specific dates.

Procedure

1. Import the required libraries (`pandas` and `matplotlib.pyplot`).
2. Read the **alphabet_stock_data.csv** file using `pd.read_csv()`.
3. Convert the **Date** column to datetime format.
4. Filter the data between the required start and end dates.
5. Create a scatter plot with **Volume** on the X-axis and **Close** price on the Y-axis.
6. Add the chart title, axis labels, and grid.
7. Display the scatter plot.

Program

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the dataset
df = pd.read_csv("alphabet_stock_data.csv")

# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# Filter data between two dates
stock_data = df[(df["Date"] >= "2020-04-01") &
            (df["Date"] <= "2020-04-30")]

# Create Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(stock_data["Volume"], stock_data["Close"], marker='o')

plt.title("Trading Volume vs Closing Stock Price")
plt.xlabel("Trading Volume")
plt.ylabel("Closing Stock Price")
plt.grid(True)
plt.show()
```

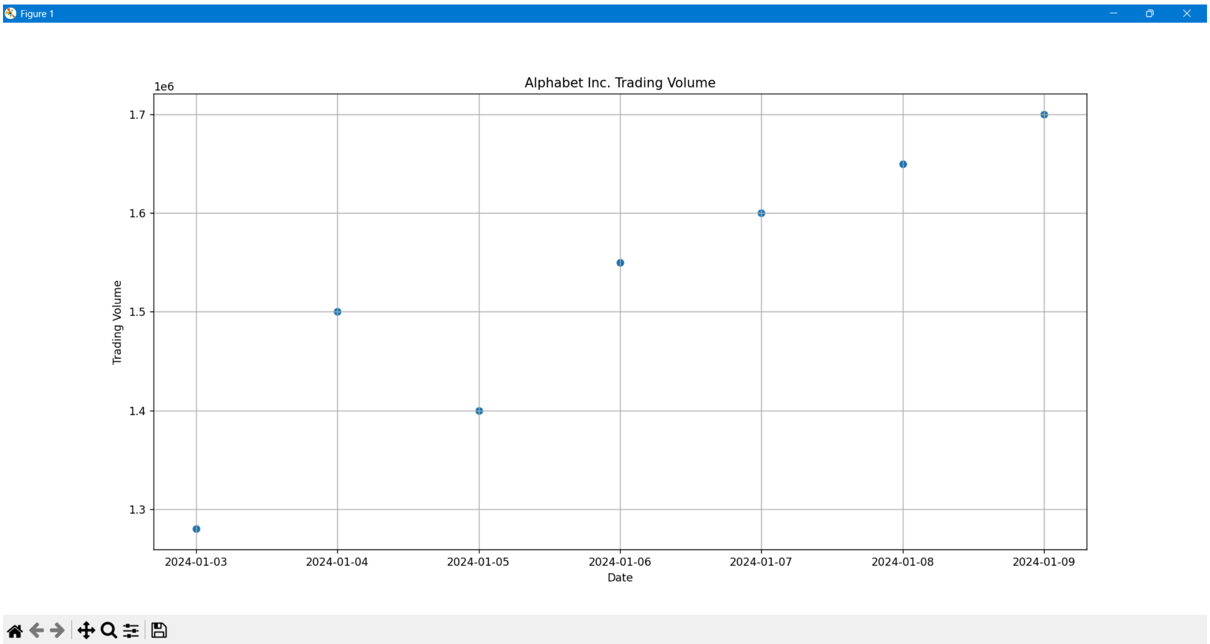
Figure 1
Alphabet Inc. Trading Volume
1e6
1.7
1.6
1.5
1.4
1.3
Trading Volume
2024-01-03
2024-01-04
2024-01-05
2024-01-06
2024-01-07
2024-01-08
2024-01-09
Date

7. Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.(refer sales_data table)

Aim

To write a Python Pandas program to create a **Pivot Table** and find the **maximum** and **minimum** sale value (`Sale_amt`) for each item in the sales data.

Procedure

1. Import the Pandas library.
2. Create or read the **sales_data.csv** file.
3. Store the data in a Pandas DataFrame.
4. Create a Pivot Table using the `pivot_table()` function.
5. Set **Item** as the index and **Sale_amt** as the values.
6. Use the aggregation functions **max** and **min**.
7. Display the Pivot Table.

Program

```
import pandas as pd

# Read the dataset
df = pd.read_csv("sales_data.csv")

# Create Pivot Table
pivot = pd.pivot_table(
    df,
    values="Sale_amt",
    index="Item",
    aggfunc=["max", "min"]
)

print("Maximum and Minimum Sale Amount for Each Item")
print(pivot)
```

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/subha/Desktop/alphabet.py =================

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==================== RESTART: C:/Users/subha/Desktop/al.py ====================

================== RESTART: C:/Users/subha/Desktop/sales.py ==================
Original Dataset:

   OrderDate   Region  Manager  ... Units Unit_price  Sale_amt
0     1-6-18     East   Martha  ...    95     1198.0    113810
1    1-23-18  Central  Hermann  ...    50      500.0     25000
2     2-9-18  Central  Hermann  ...    36     1198.0     43128
3    2-26-18  Central  Timothy  ...    27      225.0      6075
4    3-15-18     West  Timothy  ...    56     1198.0     67088
5     4-1-18     East   Martha  ...    60      500.0     30000
6    4-18-18  Central   Martha  ...    75     1198.0     89850
7     5-5-18  Central  Hermann  ...    90     1198.0    107820
8    5-22-18     West  Douglas  ...    32     1198.0     38336
9     6-8-18     East   Martha  ...    60      500.0     30000
10   6-25-18  Central  Hermann  ...    90     1198.0    107820
11   7-12-18     East   Martha  ...    29      500.0     14500
12   7-29-18     East  Douglas  ...    81      500.0     40500
13   8-15-18     East   Martha  ...    35     1198.0     41930
14    9-1-18  Central  Douglas  ...     2      125.0       250
15   9-18-18     East   Martha  ...    16       58.5       936
16   10-5-18  Central  Hermann  ...    28      500.0     14000
17  10-22-18     East   Martha  ...    64      225.0     14400

[18 rows x 8 columns]

Pivot Table (Maximum and Minimum Sale Amount):

                  max      min
             Sale_amt Sale_amt
Item
Cell Phone      14400     6075
Desk              250      250
Home Theater    40500    14000
Television     113810    38336
Video Games       936      936
>>>
Ln: 47  Col: 0
```